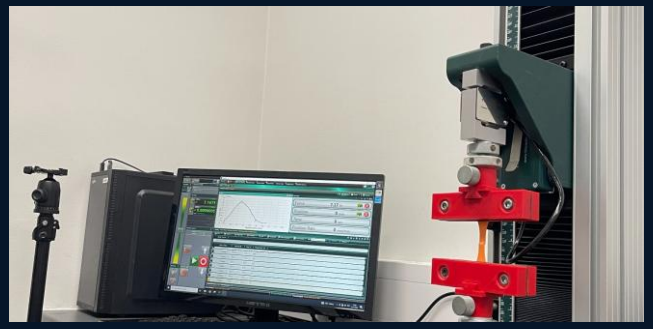
London South Bank University

ENGINEERING PORTFOLIO / 2025 – 26

MILD STEEL
TENSILE TESTING

Experimental investigation of mechanical behaviour

SOLID MECHANICS LABORATORY | MATERIALS & FRACTURE









BOKHTEAR UDDIN BADHON
BEng (Hons) Mechanical Engineering
London South Bank University

Experimental Materials Study

Solid Mechanics 2025–26

Seeking a **2027 Industrial Placement**

INNOVATIVE ENGINEERING.
A SUSTAINABLE TOMORROW.

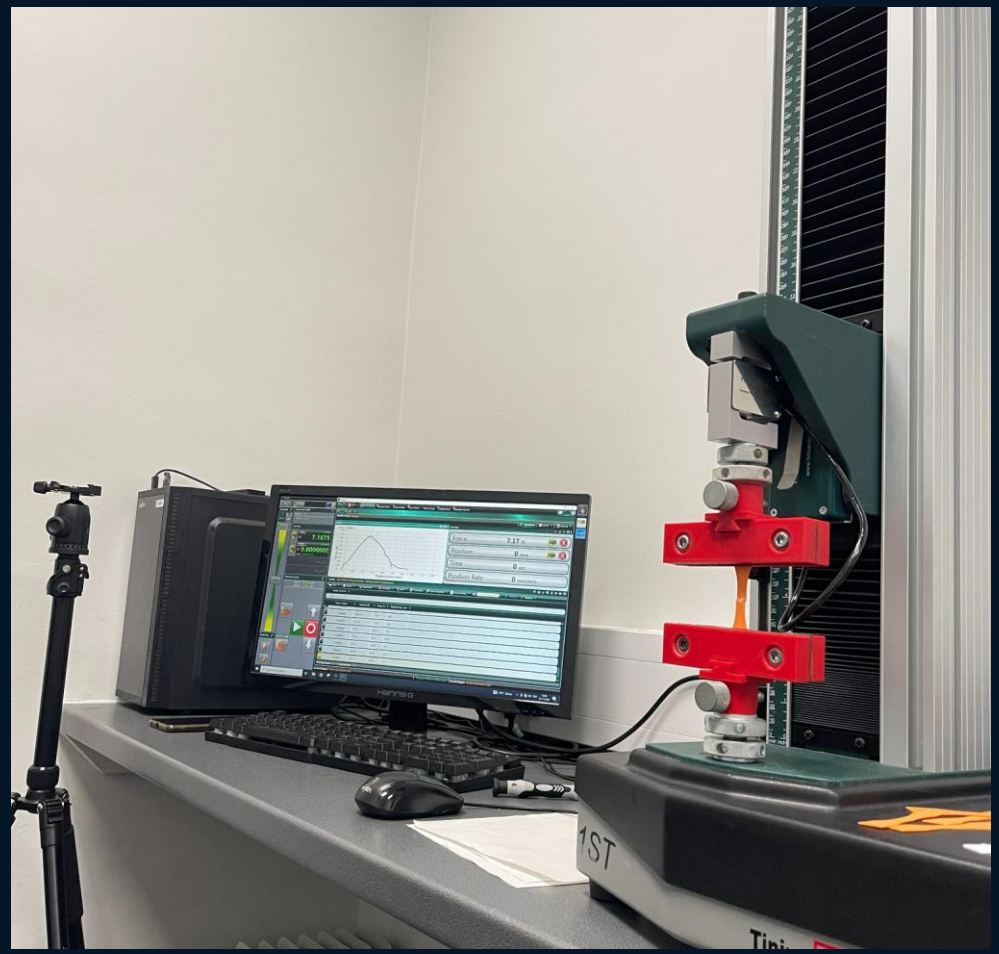
Project overview & engineering objectives

THE ENGINEERING QUESTION

How does a material respond when subjected to steadily increasing uniaxial tensile load?

EXPERIMENTAL OBJECTIVES

Identify elastic and plastic regions; examine yielding, strain hardening, necking and fracture; connect observed behaviour to material selection and safe design.



Original laboratory photography • tensile-testing system

Solid mechanics: theory into practice

ENGINEERING STRESS

$\sigma = F / A_0$

Force divided by original cross-sectional area.

Units: $N/mm^2 = MPa$.

ENGINEERING STRAIN

$\varepsilon = \Delta L / L_0$

Change in gauge length divided by original gauge length.

Dimensionless.

ELASTIC STIFFNESS

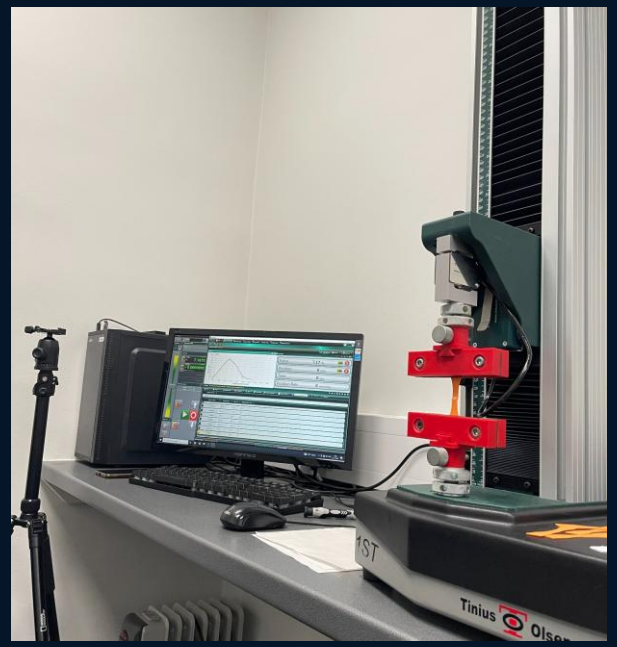
$E = \Delta\sigma / \Delta\varepsilon$

Young’s modulus is the slope of the initial linear elastic stress–strain region.

Conceptual framework from the laboratory report; numerical modulus requires suitable strain data.

Experimental setup: equipment & instrumentation





SYSTEM ELEMENTS

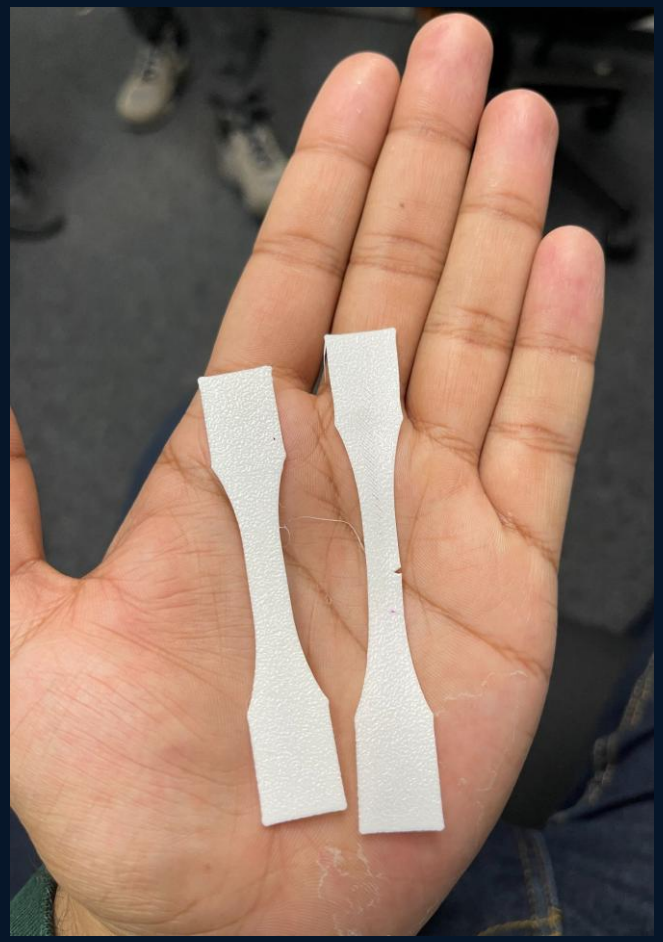
- Universal testing machine
- Upper and lower grips
- Load measurement
- Crosshead displacement
- Software data capture

Equipment identification note

The photographs show Tinius Olsen branding; the written report describes an Instron system. Confirm the actual machine model before external publication.

Specimen geometry & preparation





REPORTED DIMENSIONS

Gauge length L_0 = 50 mm

Width = 12 mm

Thickness = 6 mm

Nominal area A_0 = 72 mm^2

PRE-TEST CHECKS

Visual inspection; dimension measurement; grip alignment; load/displacement zeroing.

Experimental workflow: from setup to fracture

01 INSPECT

Check specimen surface and dimensions.

02 MOUNT

Secure the specimen and check alignment.

03 ZERO

Prepare load and extension measurement.

04 LOAD

Apply tensile displacement; record response.

05 OBSERVE

Track yielding, necking and final fracture.

06 EXPORT

Review recorded data and assess limitations.

Report-stated crosshead speed: 2 mm/min. Check machine settings against original logs.

Data acquisition: real force–displacement plots





Original screenshots show recorded force–displacement curves and specimen test rows; screen readings are not substituted for verified raw data.

Reported results & engineering interpretation

Young’s modulus	205 GPa
Upper / lower yield strength	260 / 240 MPa
Ultimate tensile strength	430 MPa
Fracture strength	380 MPa
Elongation at break	28%

RESULTS STATUS

These values are reported in the lab document as example observed mechanical properties. Raw force–extension data and a verified stress–strain calculation are needed to confirm them.

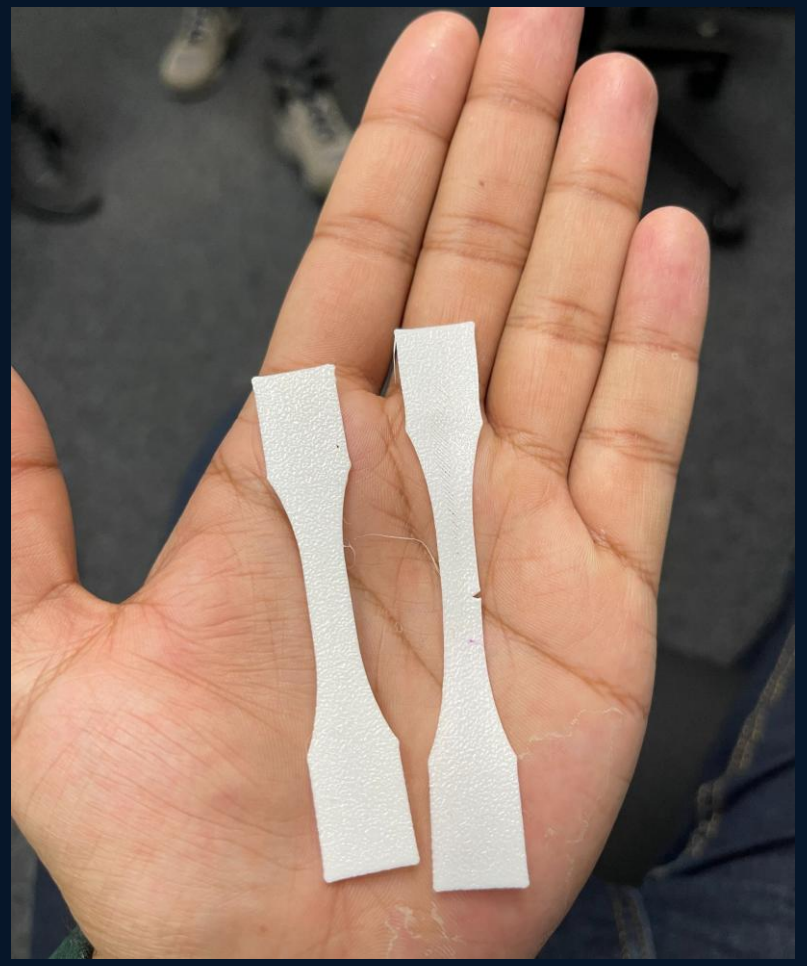
INTERPRETATION

Ductile behaviour was described in the report.

Failure behaviour: necking & fracture







Physical evidence: mounted specimen • fractured sample • original specimen geometry

Engineering discussion: what the curves show



- Initial load increase reflects specimen stiffness.
- Plastic deformation follows the elastic response.
- Strain hardening increases load resistance.
- Necking localises deformation before fracture.
- Force–displacement is not automatically stress–strain.

Critical evaluation & practical improvements

POTENTIAL UNCERTAINTIES

- Misalignment and bending strains
- Dimensional measurement uncertainty
- Extensometer removal and strain accuracy
- Grip slip or crosshead compliance
- Limited repeatability without replicate tests

RECOMMENDED IMPROVEMENTS

- Verify machine, test speed and specimen ID
- Repeat tests and report variation
- Use suitable strain instrumentation
- Check grip alignment and calibration
- Retain and plot original numerical data

Engineering reflection & 2027 placement goal

WHAT THIS PROJECT DEMONSTRATES

Applied understanding of tensile behaviour, specimen geometry, laboratory instrumentation, experimental data interpretation and evidence-based technical reporting.

NEXT STEP: INDUSTRIAL PRACTICE

I am developing my practical engineering judgement and communication skills through first-year BEng Mechanical Engineering projects, while preparing for a 2027 industrial placement.



BOKHTEAR UDDIN BADHON